

$$L \frac{di_L}{dt} + Ri_L = U_s$$

$$LP + R = 0$$

$$P = -\frac{R}{L}$$

$$i_L = i_L' + i_L''$$

特解为常数

$$i_L' = \frac{U_s}{R}$$

$$= \frac{U_s}{R} + Ae^{-\frac{R}{L}t}$$

$$i_L(0^+) = \frac{U_s}{R} + A = 0$$

$$A = -\frac{U_s}{R}$$

$$i_L = \frac{U_s}{R} (1 - e^{-\frac{R}{L}t})$$

$$u_L = L \frac{di_L}{dt} = L \cdot \frac{U_s}{R} \cdot \left(-\frac{R}{L}\right) \cdot -e^{-\frac{R}{L}t}$$

$$= U_s e^{-\frac{R}{L}t}$$

